JESTER N. ITLIONG

Ph.D. Candidate in Physics (expected July 2026)

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Houghton, MI - 49931, USA

Professional Summary

Ph.D. Candidate in Physics with expertise in multiscale molecular simulations of polymers, ionic liquids, and soft materials. Experienced in atomistic and coarse-grained molecular dynamics simulations, statistical mechanics, and high-performance scientific computing to investigate transport, rheological, dielectric, and structural properties of complex polymeric systems. Research focuses on connecting molecular interactions, mesoscale organization, and macroscopic material behavior through scalable computational workflows and physics-based modeling. Strong background in polymer physics, transport phenomena, and structure–property relationships with over 8 years of interdisciplinary computational research experience.

Research Interests

- Multiscale Modeling of Polymers and Soft Materials
- Coarse-Grained Molecular Dynamics
- Structure-Property Relationships
- Polymer Transport Phenomena
- Rheology and Relaxation Dynamics
- Mesoscale Organization in Polymeric Systems
- Statistical Mechanics of Soft Matter
- Physics-Based Computational Materials Modeling

Education

- **Michigan Technological University** — *January 2021 - present*
  PhD in Physics — Houghton, MI, USA
  - Dissertation: Molecular Origins of Viscosity in Ionic and Polymerized Ionic Liquids: Insights from Stockmayer Fluid Simulations
  - Focus: coarse-grained MD simulations, polymer physics,multiscale transport phenomena, ionic and polymerized ionic liquids, rheology and structure–property relationships, mesoscale organization in charged polymer systems
  - Advisor: Prof. Issei Nakamura
  - GPA: 4.00/4.00
- **De La Salle University** — *December 2017*
  Master of Science in Physics — Manila, Philippines
  - Thesis: Molecular Dynamics Study of Water Transport and Reverse Solute Diffusion Mechanisms across a Polyamide Membrane during Forward-osmosis-driven Dewatering of Microalgae
  - Focus: atomistic MD simulations, polymer and lipid membranes, ion and water transport, forward osmosis
  - Advisor: Prof. Nelson B. Arboleda, Jr.
  - GPA: 3.15/4.00
- **Eulogio "Amang" Rodriguez Institute of Science and Technology** — *March 2014*
  Bachelor of Science in Applied Physics with minor in Computer Science — Manila, Philippines
  - Thesis: Neutron Flux Measurements inside the Am-Be Neutron Calibrator Source
  - Focus: neutron physics, neutron radiation
  - GPA: 1.60/1.00 (Cum Laude)

RESEARCH EXPERIENCE

- **Michigan Technological University** *January 2021 - present*
  Graduate Research Assistant Houghton, MI, USA
  - Develop coarse-grained and molecular-scale simulation models for ionic liquids and polymerized ionic liquid systems to investigate transport, rheological, dielectric, and structural behavior in complex polymeric materials.
  - Perform equilibrium and non-equilibrium molecular dynamics simulations using LAMMPS to study chain dynamics, ionic transport, stress relaxation, and structure–property relationships in charged polymer systems.
  - Apply Green–Kubo, Einstein–Helfand, and correlation-function-based approaches to analyze viscosity, ionic conductivity, relaxation dynamics, and collective transport phenomena.
  - Design scalable Python-based computational workflows for automated simulation analysis, data fitting, and large-scale statistical-mechanical calculations.
  - Develop physically informed coarse-grained frameworks that extend accessible simulation time and length scales for polymer systems.
  - Integrate simulation results with experimental observations to validate molecular-level transport and rheological behavior.
  - Collaborate with researchers from Sandia National Laboratories on computational modeling and simulation methodologies.
  - Mentor undergraduate and graduate researchers in molecular simulation methods, scientific programming, and computational data analysis..
- **De La Salle University** *April 2018 - December 2020*
  Research Assistant Manila, Philippines
  - Performed atomistic molecular dynamics simulations of polymer, membrane, and electrolyte systems to investigate water and solute transport mechanisms in soft materials.
  - Developed force-field parameters for polymer systems using density functional theory calculations.
  - Investigated interfacial transport behavior and molecular interactions in polymer–solution environments.
  - Collaborated within interdisciplinary research teams combining computational modeling, materials science, and transport analysis.
  - Published peer-reviewed research articles and presented findings at local and international conferences.

TEACHING EXPERIENCE

- **De La Salle University** *January 2018 - December 2020*
  Assistant Professorial Lecturer (Part-time) Manila, Philippines
- **Eulogio "Amang" Rodriguez Institute of Science and Technology** *November 2017 - December 2020*
  Instructor (Full-time) Manila, Philippines

TECHNICAL SKILLS

- **Molecular Simulation and Modeling:** Molecular Dynamics Simulations, Coarse-Grained Modeling, Statistical Mechanics, Transport Phenomena, Polymer Physics, Soft-Matter Modeling, Rheology and Viscosity Calculations, Monte Carlo Simulations
- **Scientific Computing and HPC:** Python, C++, Fortran, MPI/OpenMP, SLURM, Linux, High-Performance Computing)
- **Simulation Software:** LAMMPS, GROMACS, OVITO, VMD, Gaussian, Avogadro
- **Data Science and Analysis:** NumPy, SciPy, pandas, matplotlib, Machine Learning (introductory), Scientific Data Analysis, Numerical Modeling

HONORS AND AWARDS

- **Finalist** *March 2026*
  Regional (Midwest) 3-Minute Thesis Competition
  - Represented Michigan Tech and advanced to the final round with 9 other finalists in the 3-Minute Thesis Competition organized by the Midwestern Association of Graduate Schools in Kansas City, MO.
- **1st Place** *November 2025*
  3-Minute Thesis Competition
  - Won 1st place over 12 other finalists in the 3-Minute Thesis Competition organized by the Graduate Student Government of Michigan Tech.
- **Best T-shirt Design** *March 2025*
  APS Division of Polymer Physics (DPOLY)
  - Won the best design for the official t-shirt souvenir of DPOLY during the APS March 2025 Meeting.
- **Best Poster Presentation** *April 2023*
  Graduate Physics Colloquium at Michigan Tech

- Awarded as Best Poster for my presentation "Effects of Electric Charge, Dipole Moment, and Molecular Size on the Viscosity of Ionic Liquids".

- **Outstanding Faculty** *October 2019*
  Faculty Recognition at Eulogio "Amang" Rodriguez Institute of Science and Technology
  - Awarded as one of the Outstanding Faculty in the College of Arts and Sciences. The award was voted by over 100 undergraduate students.

Leadership and Service

- **Volunteer** *March 16-21, 2025*
  APS Division of Polymer Physics
  - Assisted in managing the membership table of DPOLY during the APS March 2025 Meeting.
- **Resource Speaker** *January 24-26, 2025*
  APS Conference for Undergraduate Women and Gender Minorities in Physics at Michigan Tech
  - Presented and shared my experiences and insights as a senior graduate student and as someone pursuing a career in education and research.
- **Graduate Physics Representative** *May 2023 - April 2024*
  Graduate Student Government at Michigan Tech
  - Advocated for graduate students' concerns and welfare as the Physics Dept. representative in the Graduate Student Government.
  - Developed and implemented a new peer evaluation procedure for Graduate Physics Colloquium.
  - Participated in student body and town hall meetings to review and provide feedback on proposed student policies.
- **Research Coordinator** *June 2018 - December 2020*
  College of Arts and Sciences, Eulogio "Amang" Rodriguez Institute of Science and Technology
  - Organized institution-wide research colloquia.
  - Participated meetings for revising research guidelines and policies.
  - Developed a systematic archiving workflow for thesis and research projects of students.

Professional Memberships

- **American Physical Society - Division of Polymer Physics** *October 2021 - Present*
- **Women in Physics at Michigan Tech** *August 2021 - Present*
- **Philippine Association of Physics and Science Instructors** *January 2015 - December 2020*

Publications

C=Conference, J=Journal

Polymerized Ionic Liquids and Polymer Dynamics

[J.4] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I., (2026). **Viscosity, Chain Dynamics, and Ion-pairing in Polymerized Ionic Liquids: Effects of Cation Size and Dipole Mobility**. *In preparation.*

[J.2] Gao, T., Itliong J., Kumar S. P., Hjorth Z., and Nakamura, I., (2021). **Polarization of Ionic Liquid and Polymer and its Implications for Polymerized Ionic Liquids: An Overview Towards a New Theory and Simulation**. *Journal of Polymer Science*, Vol. 59, Issue 21, pp. 2434-2457. DOI: 10.1002/pol.20210330

Ionic Liquid Transport and Viscosity

[J.3] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I., (2025). **Stockmayer Fluid Simulations for Viscosity and Glass Transition Temperature of Ionic Liquids**. *The Journal of Chemical Physics*, Vol. 163, Issue 4, pp. 044503. DOI: 10.1063/5.0268727

Polymer Membrane and Ion Diffusion

[J.1] Itliong, J.N., Villagracia, A.R.C., Moreno, J.L.V., et. al. (2019). **Investigation of reverse ionic diffusion in forward-osmosis-aided dewatering of microalgae: A molecular dynamics study**. *Bioresource Technology*, Vol. 279, pp. 181-188. DOI: 10.1016/j.biortech.2019.01.109

[C.2] Itliong, J.N., Villagracia, A.R.C., Ong, H.L., et. al. (2019). **H2O Absorptivity on a Fully 4-crosslinked Polyacrylamide Membrane via Density Functional Theory and Monte Carlo Calculations for Draw Solution Recovery in Forward Osmosis**. In *2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, pp. 1-5. Laoag, Philippines. DOI: 10.1109/HNICEM48295.2019.9072782

[C.1] Itliong, J.N., Villagracia, A.R.C., Moreno, J.L.V., et. al. (2018). **Density Functional Theory-based modeling and calculations of a polyamide molecular unit for studying forward-osmosis-dewatering of microalgae**. In *2018 IEEE 10th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, pp. 1-5. Baguio City, Philippines. DOI: 10.1109/HNICEM.2018.8666436.

Delivered and Contributed Presentations

[7] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I. "Role of Cation Size and Dipolar Motion Restriction in the Viscosity of Polymerized Ionic Liquids". American Physical Society Global Physics Summit, Denver, CO, March 2026.

[6] Nakamura, I, Itliong, J.N., Frischknecht, A.L., and Stevens, M.J. "Polymerized Stockmayer Fluid MD Simulation of Ion Transport in Polymeric Ionic Liquid–LiTFSI Blends". American Physical Society Global Physics Summit, Denver, CO, March 2026.

[5] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I. "Effects of Molecular Size and Polarity on Viscosity Scaling and Ionic Conductivity of Polymerized Ionic Liquids". American Physical Society Global Physics Summit, San Diego, CA, March 2025.

[4] Nakamura, I., Itliong, J.N., Karkhanis, S.S., Stevens, M.J., and Frischknecht, A.L. "Contributions of Rotational and Collective Translational Dipole Moments to Dielectric Constants in Polymeric Ionic Liquids: A Coarse-Grained Stockmayer Molecular Dynamics Study". American Physical Society Global Physics Summit, San Diego, CA, March 2025.

[3] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I. "Influences of Chain Polarity and Molecular Weight on Ion and Polymer Dynamics in Polymerized Ionic Liquids". American Physical Society March Meeting, Minneapolis, MN, March 2024.

[2] Nakamura, I., Shock, C.J., Itliong, J.N., Stevens, M.J., and Frischknecht, A.L. "Effects of Ion Solvation in a Mixture of Polar Solvent and Polymer". American Physical Society March Meeting, Minneapolis, MN, March 2024.

[1] Itliong, J.N., Frischknecht, A.L., Stevens, M.J., and Nakamura, I. "Effects of Electric Charge, Dipole Moment, and Molecular Size on the Viscosity of Ionic Liquids". American Physical Society March Meeting, Las Vegas, NV, March 2023.

References

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